

MOTION AND DIFFERENCE

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Abstract

We construct one explicit mathematical realization in which a two-dimensional carrier supports complex phase, square symmetry, projective arithmetic reflection, Fibonacci refinement, and a Lorentzian $SO(2)$ gauge field. The elementary factorization $Q = M_J C_1 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ turns alternating projective involutions into the exact matrix record of the Fibonacci sequence. Its two fixed points form a maximal equalizer in $PGL(2, \mathbb{R})$, determine a hyperbolic glide axis, and recover the sharp Diophantine scale $\sqrt{5}$. On a Lorentzian four-manifold, the same phase generator C_1 carries a global $SO(2)$ connection. A nonnegative gluing functional detects exact compatibility of local bundle data. An explicit one-parameter family of connections keeps a nonzero source-free Maxwell curvature fixed while varying relative holonomy independently; a pair of endpoint-sharing timelike paths also has unequal proper time. The completed zeta reflection is placed on the same arithmetic involution, with no asserted spectral, causal, or functional identification among the resulting scalar and geometric quantities. The main result is therefore a compatibility theorem: the declared structures coexist exactly in a single realization, while the boundaries between them remain explicit.

Keywords: projective dynamics; Fibonacci matrix; hyperbolic geometry; Diophantine approximation; Lorentzian gauge field; holonomy; Maxwell equations.

Classification: 11B39, 11J70, 20H10, 53C29, 58J28, 83C50.

I. STATEMENT AND SCOPE

The object of this paper is narrow. We ask whether several exact structures, normally written in separate mathematical languages, can be carried without contradiction by one explicit model. The answer is affirmative. The model contains a common two-dimensional carrier, but it does not collapse the roles placed on that carrier.

The arithmetic core begins with two involutions on the real projective line. One is induced by a complex quarter-turn; the other is induced by the affine reflection $x \mapsto 1 - x$. Their alternating composition is represented by the classical Fibonacci matrix. The same reflection is also the argument symmetry of the completed Riemann zeta function. These statements meet at a precisely declared two-point set and nowhere else.

The geometric core begins with an oriented rank-two Euclidean bundle over a Lorentzian four-manifold. Its $SO(2)$ connection uses the same infinitesimal quarter-turn generator. Exact gluing is detected by a positive functional. In an explicit product spacetime, the connection has nonzero source-free Maxwell curvature, independently variable holonomy, and timelike paths of unequal proper time.

The classical ingredients concerning Fibonacci numbers, continued fractions, Hurwitz approximation, hyperbolic isometries, the completed zeta function, connections, and Maxwell fields are standard. The point proved here is their exact placement in a single bounded construction, centered on the factorization $Q = M_J C_1$ and completed by an explicit Lorentzian witness.

Theorem 1 (Simultaneous realization). *There exists a tuple*

$$\mathfrak{A} = (V, C_1, \Pi, M_J, Q, J, \xi, M, g, E, \mathcal{A}, q, \gamma_0, \gamma_1)$$

with the following properties.

- (i) $\langle C_1, \Pi \rangle \cong D_4$ on $V = \mathbb{R}^2$ and $\mathbb{P}(D_4) \cong V_4$;
- (ii) $Q = M_J C_1$, $Q^2 = Q + I$, and the alternating projective iterate has Fibonacci coefficients;
- (iii) the equalizer of the two projective involutions is maximal and consists exactly of the two roots of $s^2 - s - 1$;
- (iv) Q acts on the upper half-plane as a glide reflection of displacement $2 \log \Phi$, and the Fibonacci convergents attain the sharp Hurwitz scale asymptotically;
- (v) the completed zeta function obeys $\xi \circ J = \xi$, including equality on the two refinement roots;
- (vi) $E \rightarrow M$ carries an $SO(2)$ connection with nonzero source-free Maxwell curvature and exact local gluing;
- (vii) within a one-parameter family, curvature is fixed while relative holonomy is arbitrarily prescribable in $SO(2)$;
- (viii) two timelike paths with common endpoints have unequal proper time.

No equality or functional dependence is asserted among the zeta remainder, Maxwell curvature, relative holonomy, proper-time difference, and gluing functional.

The proof occupies Sections II–VIII. Each interface is typed before it is used, and the nonidentifications are collected in Section IX.

II. THE CARRIER AND THE ALTERNATING FACTORIZATION

Let $V = \mathbb{R}^2$ with its standard Euclidean inner product. We use the same real space as $\mathbb{C}$ for phase and as $\mathbb{P}(V) \cong \mathbb{RP}^1$ for projective action. Define

$$C_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \Pi = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad M_J = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}. \quad (1)$$

Then

$$C_1^2 = -I, \quad C_1^4 = I, \quad \Pi^2 = I, \quad \Pi C_1 \Pi = C_1^{-1}, \quad M_J^2 = I. \quad (2)$$

Thus $\langle C_1, \Pi \rangle \cong D_4$, the order-eight symmetry group of the square. Under $V \cong \mathbb{C}$, multiplication by i is C_1 , and

$$e^{\theta C_1} = I \cos \theta + C_1 \sin \theta, \quad e^{\pi C_1} = -I, \quad e^{2\pi C_1} = I. \quad (3)$$

The projective image has kernel $\{I, -I\}$, hence

$$\mathbb{P}(D_4) \cong D_4 / \{\pm I\} \cong V_4. \quad (4)$$

Let $\mathbb{P}(A)$ denote the projective map induced by an invertible real matrix A . Put

$$\sigma = \mathbb{P}(C_1), \quad J_{\mathbb{P}} = \mathbb{P}(M_J). \quad (5)$$

On the affine chart $\iota(x) = [x : 1]$, these are

$$\sigma_{\text{aff}}(x) = -\frac{1}{x}, \quad J_{\text{aff}}(x) = 1 - x. \quad (6)$$

Both projective maps are involutions. They are distinct because the signs of $\det C_1 = 1$ and $\det M_J = -1$ cannot be changed by rescaling a real matrix representative.

The central factorization is

$$Q = M_J C_1 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}. \quad (7)$$

Its characteristic polynomial is $\lambda^2 - \lambda - 1$. Cayley-Hamilton gives

$$Q^2 = Q + I, \quad Q^{-1} = Q - I = C_1^{-1} M_J. \quad (8)$$

Defining $\mathcal{T} = \mathbb{P}(Q)$, we obtain

$$\mathcal{T} = J_{\mathbb{P}} \circ \sigma, \quad \mathcal{T}^{-1} = \sigma \circ J_{\mathbb{P}}. \quad (9)$$

On the affine chart,

$$T(x) = 1 + \frac{1}{x}, \quad T^{-1}(x) = \frac{1}{x-1}. \quad (10)$$

The order matters. Directly,

$$[Q, C_1] = QC_1 - C_1Q = \begin{pmatrix} 2 & -1 \\ -1 & -2 \end{pmatrix} \neq 0. \quad (11)$$

2.1. The maximal meeting

Let $P(s) = s^2 - s - 1$ and define the affine reflection

$$J(s) = 1 - s. \quad (12)$$

The polynomial is invariant under J :

$$P \circ J = P. \quad (13)$$

Its roots are

$$\Phi = \frac{1 + \sqrt{5}}{2}, \quad J(\Phi) = \frac{1 - \sqrt{5}}{2} = -\frac{1}{\Phi}. \quad (14)$$

Equivalently, a whole-remainder proportion $W = L + R$ and $W/L = L/R = x$ yields $x = T(x)$, hence $P(x) = 0$. In the centered coordinate $w = s - 1/2$, the same equation is $w^2 = 5/4$.

Proposition 2 (Maximal equalizer). *Let $\mathcal{R}_{\Phi} = \{\Phi, J(\Phi)\}$. Then*

$$\text{Eq}(J_{\mathbb{P}}, \sigma) = \text{Fix}(\mathcal{T}) = \text{Fix}(\mathcal{T}^{-1}) = \{[x : 1] : x \in \mathcal{R}_{\Phi}\}. \quad (15)$$

This two-point set is the largest possible equalizer of two distinct elements of $\text{PGL}(2, \mathbb{R})$.

Proof. Since $J_{\mathbb{P}}$ is an involution, $J_{\mathbb{P}}(p) = \sigma(p)$ if and only if $(J_{\mathbb{P}} \circ \sigma)(p) = p$. Neither 0 nor ∞ is fixed by $\mathcal{T}$. On the affine chart, $T(x) = x$ if and only if $P(x) = 0$, giving the two points in Eq. (15). The inverse has the same fixed points. For distinct $f, g \in \text{PGL}(2, \mathbb{R})$, $\text{Eq}(f, g) = \text{Fix}(g^{-1}f)$. A nonidentity real projective transformation has at most two real eigenlines, hence at most two fixed points in $\mathbb{RP}^1$. $\square$

Corollary 3. *The group generated by the two projective involutions is infinite dihedral:*

$$\langle J_{\mathbb{P}}, \sigma \rangle \cong D_{\infty}. \quad (16)$$

Proof. The eigenvalues of Q are Φ and $-\Phi^{-1}$. Their ratio has modulus $\Phi^2 \neq 1$, so $\mathcal{T} = \mathbb{P}(Q)$ has infinite order. Two involutions whose product has infinite order generate D_{∞} . $\square$

III. FIBONACCI DYNAMICS AND THE HYPERBOLIC AXIS

Let $F_0 = 0$, $F_1 = 1$, and $F_{n+1} = F_n + F_{n-1}$ for $n \geq 1$.

Proposition 4 (Fibonacci powers). *For every $n \geq 1$,*

$$Q^n = F_n Q + F_{n-1} I = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}. \quad (17)$$

Consequently,

$$(J_{\mathbb{P}} \circ \sigma)^n = \mathbb{P}(Q^n), \quad \mathcal{T}^n(\infty) = \frac{F_{n+1}}{F_n}. \quad (18)$$

Proof. The assertion holds for $n = 1$. If $Q^n = F_n Q + F_{n-1} I$, then Eq. (8) gives

$$Q^{n+1} = F_n(Q + I) + F_{n-1}Q = F_{n+1}Q + F_n I.$$

The matrix formula and projective statements follow. $\square$

The ratios in Eq. (18) are the finite continued fractions $[1; 1, \dots, 1]$ with n entries. The exact Binet normalization is

$$F_n = \frac{\Phi^n - J(\Phi)^n}{\Phi - J(\Phi)} = \frac{\Phi^n - J(\Phi)^n}{\sqrt{5}}, \quad (19)$$

and

$$F_{n+1} - \Phi F_n = J(\Phi)^n. \quad (20)$$

It follows that

$$\sqrt{5} F_n^2 \left| \Phi - \frac{F_{n+1}}{F_n} \right| = 1 - (-1)^n \Phi^{-2n}, \quad (21)$$

so

$$\lim_{n \rightarrow \infty} \sqrt{5} F_n^2 \left| \Phi - \frac{F_{n+1}}{F_n} \right| = 1. \quad (22)$$

Hurwitz's theorem [3, 4] states that every irrational α has infinitely many rational approximants p/q with

$$\left| \alpha - \frac{p}{q} \right| < \frac{1}{\sqrt{5} q^2}, \quad (23)$$

and that the constant is optimal uniformly over irrational α . Equation (22) places the Fibonacci convergents to Φ at that sharp asymptotic scale. The same $\sqrt{5}$ is also the exact separation

$$\Phi - J(\Phi) = \sqrt{5} \quad (24)$$

of the two refinement branches. These are two declared mathematical offices of one constant, not a causal relation.

Taking determinants in Eq. (17) yields Cassini's identity,

$$F_{n+1} F_{n-1} - F_n^2 = (-1)^n. \quad (25)$$

Thus even projective iterates preserve orientation and odd iterates reverse it.

3.1. Upper-half-plane realization

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL(2, \mathbb{R})$, define the standard extended action on $\mathbb{H}$ by Refs. [2, 6]:

$$\rho_A(z) = \begin{cases} (az + b)/(cz + d), & \det A > 0, \\ (a\bar{z} + b)/(c\bar{z} + d), & \det A < 0. \end{cases} \quad (26)$$

Then

$$\rho_{C_1}(z) = -\frac{1}{z}, \quad \rho_{M_1}(z) = 1 - \bar{z}, \quad \rho_Q(z) = 1 + \frac{1}{\bar{z}}. \quad (27)$$

The first is rotation by π about i . The second is reflection in $\operatorname{Re} z = 1/2$. The third preserves the geodesic with endpoints $J(\Phi)$ and Φ ,

$$\left| z - \frac{1}{2} \right| = \frac{\sqrt{5}}{2}, \quad \operatorname{Im} z > 0. \quad (28)$$

Proposition 5 (Glide displacement). *The map ρ_Q is an orientation-reversing glide reflection along the axis (28), with glide displacement $2 \log \Phi$. Its square is a hyperbolic translation of length $4 \log \Phi$.*

Proof. The boundary fixed points solve $x = 1 + 1/x$, hence are the two points of $\mathcal{R}_\Phi$. The map has no fixed point in $\mathbb{H}$, so it is a glide reflection along their geodesic. Moreover,

$$Q^2 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \in SL(2, \mathbb{R}), \quad \operatorname{tr}(Q^2) = 3,$$

with eigenvalues Φ^2 and Φ^{-2} . Its translation length is

$$2 \operatorname{arccosh}(3/2) = 4 \log \Phi. \quad (29)$$

A glide reflection squares to translation by twice its displacement. $\square$

Equations (17) and (27) therefore describe one discrete orbit in three compatible readings: integer matrix coefficients, continued-fraction refinement, and alternating translation-reflection along a fixed hyperbolic axis.

IV. THE COMPLETED-ZETA REFLECTION

Let ζ_R denote the analytically continued Riemann zeta function and define

$$\tilde{\zeta}(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta_R(s). \quad (30)$$

The function $\tilde{\zeta}$ is entire and satisfies the classical functional equation [8]

$$\tilde{\zeta}(s) = \tilde{\zeta}(1-s), \quad (31)$$

or, using Eq. (12),

$$\tilde{\zeta} \circ J = \tilde{\zeta}. \quad (32)$$

Since the refinement roots form a J -pair,

$$\tilde{\zeta}(\Phi) = \tilde{\zeta}(J(\Phi)). \quad (33)$$

Also, because σ_{aff} exchanges these same two points,

$$(\tilde{\zeta} \circ \sigma_{\text{aff}})|_{\mathcal{R}_\Phi} = \tilde{\zeta}|_{\mathcal{R}_\Phi}. \quad (34)$$

Equation (34) is a two-point restriction, not a global identity on $\mathbb{R}^\times$.

For the antiholomorphic reflection $J_{\mathbb{H}}(s) = 1 - \bar{s}$, Schwarz reflection and Eq. (31) give

$$\tilde{\zeta}(J_{\mathbb{H}}(s)) = \overline{\tilde{\zeta}(s)}, \quad |\tilde{\zeta}(J_{\mathbb{H}}(s))| = |\tilde{\zeta}(s)|. \quad (35)$$

This is the standard symmetry about the critical line. Nothing here implies a claim about the location of zeta zeros.

For later separation of types, define only the scalar remainder

$$\epsilon_\Phi = \zeta_R(\Phi) - \sqrt{5}. \quad (36)$$

Then $\zeta_R(\Phi) = \sqrt{5} + \epsilon_\Phi$ is an identity by definition. No magnitude, geometric meaning, or coupling is assigned to ϵ_Φ .

V. LORENTZIAN BUNDLE STRUCTURE AND EXACT GLUING

Let (M, g) be a time-oriented Lorentzian four-manifold and let h be an auxiliary Riemannian metric on M . Let $\pi : E \rightarrow M$ be an oriented rank-two real vector bundle with structure group $SO(2)$ and positive-definite fiber metric h_E . On a finite trivializing cover $\{U_\alpha\}$, take local section representatives $q_\alpha : U_\alpha \rightarrow V$, transition maps $\mathcal{U}_{\alpha\beta} : U_{\alpha\beta} \rightarrow SO(2)$, and local connection one-forms $\mathcal{A}_\alpha \in \Omega^1(U_\alpha)$; see Refs. [5, 7, 1].

Normalize

$$\mathcal{U}_{\alpha\alpha} = I, \quad \mathcal{U}_{\beta\alpha} = \mathcal{U}_{\alpha\beta}^{-1}, \quad (37)$$

and choose overlaps on which

$$\mathcal{U}_{\alpha\beta} = e^{\chi_{\alpha\beta} C_1} \quad (38)$$

for smooth real lifts $\chi_{\alpha\beta}$. Different lifts differ locally by $2\pi\mathbb{Z}$, so $d\chi_{\alpha\beta}$ is lift-independent. Define

$$D_\alpha q_\alpha = dq_\alpha + \mathcal{A}_\alpha C_1 q_\alpha. \quad (39)$$

The connection law

$$\mathcal{A}_\beta = \mathcal{A}_\alpha - d\chi_{\alpha\beta} \quad (40)$$

makes $F_\alpha = d\mathcal{A}_\alpha$ global. Because $\mathfrak{so}(2) = \mathbb{R}C_1$, the matrix-valued connection and curvature are

$$\hat{\mathcal{A}}_\alpha = \mathcal{A}_\alpha C_1, \quad \Omega_\alpha = d\hat{\mathcal{A}}_\alpha + \hat{\mathcal{A}}_\alpha \wedge \hat{\mathcal{A}}_\alpha = F_\alpha C_1. \quad (41)$$

The Abelian quadratic term vanishes. Thus C_1 is at once the complex phase generator, the quarter-turn in D_4 , the infinitesimal generator of parallel transport, and the Lie-algebra value of curvature. This is shared carrier structure. It is not equality of the scalar quantities multiplying C_1 .

The arithmetic reflection remains outside the gauge group. Indeed,

$$M_J^T M_J = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \neq I, \quad M_J \notin O(2). \quad (42)$$

Hence M_J is neither an $SO(2)$ structure transformation nor a fiber isometry.

5.1. A zero-detecting functional

Define the local defects

$$\Delta_{\alpha\beta}^q = q_\beta - \mathcal{U}_{\alpha\beta} q_\alpha, \quad (43)$$

$$\Delta_{\alpha\beta}^{\mathcal{A}} = \mathcal{A}_\beta - \mathcal{A}_\alpha + d\chi_{\alpha\beta}, \quad (44)$$

$$\Delta_{\alpha\beta\gamma}^{\mathcal{U}} = \mathcal{U}_{\alpha\beta}\mathcal{U}_{\beta\gamma}\mathcal{U}_{\gamma\alpha} - I. \quad (45)$$

Assume continuity and square-integrability with respect to h and h_E . For fixed $\kappa_{\mathcal{A}}, \kappa_{\mathcal{U}} > 0$, set

$$\mathcal{F}_q = \sum_{\alpha < \beta} \int_{U_{\alpha\beta}} \|\Delta_{\alpha\beta}^q\|_{h_E}^2 d\text{vol}_h, \quad (46)$$

$$\mathcal{F}_{\mathcal{A}} = \sum_{\alpha < \beta} \int_{U_{\alpha\beta}} \|\Delta_{\alpha\beta}^{\mathcal{A}}\|_h^2 d\text{vol}_h, \quad (47)$$

$$\mathcal{F}_{\mathcal{U}} = \sum_{\substack{\alpha < \beta < \gamma \\ U_{\alpha\beta\gamma} \neq \emptyset}} \int_{U_{\alpha\beta\gamma}} \|\Delta_{\alpha\beta\gamma}^{\mathcal{U}}\|_{\text{HS}}^2 d\text{vol}_h, \quad (48)$$

$$\mathcal{F}_{\Lambda} = \mathcal{F}_q + \kappa_{\mathcal{A}}\mathcal{F}_{\mathcal{A}} + \kappa_{\mathcal{U}}\mathcal{F}_{\mathcal{U}}. \quad (49)$$

Proposition 6 (Exact gluing). *Under the stated assumptions,*

$$\mathcal{F}_{\Lambda} = 0 \iff \Delta_{\alpha\beta}^q = 0, \quad \Delta_{\alpha\beta}^{\mathcal{A}} = 0, \quad \Delta_{\alpha\beta\gamma}^{\mathcal{U}} = 0 \quad (50)$$

on every relevant overlap.

Proof. Every integrand in Eq. (49) is nonnegative. If all defects vanish, the functional vanishes. Conversely, a zero finite sum with positive coefficients forces each integral to vanish. If a continuous squared norm were positive at one point, it would remain positive on a nonempty open neighborhood of positive h -volume, a contradiction. $\square$

The auxiliary metric changes the numerical value of $\mathcal{F}_{\Lambda}$ but not its zero locus.

5.2. Mirror covariance

Under

$$q_\alpha \mapsto \Pi q_\alpha, \quad \mathcal{A}_\alpha \mapsto -\mathcal{A}_\alpha, \quad \chi_{\alpha\beta} \mapsto -\chi_{\alpha\beta}, \quad \mathcal{U}_{\alpha\beta} \mapsto \Pi \mathcal{U}_{\alpha\beta} \Pi, \quad (51)$$

the defects transform as

$$\Delta^q \mapsto \Pi \Delta^q, \quad \Delta^{\mathcal{A}} \mapsto -\Delta^{\mathcal{A}}, \quad \Delta^{\mathcal{U}} \mapsto \Pi \Delta^{\mathcal{U}} \Pi. \quad (52)$$

Orthogonality of Π therefore gives

$$\mathcal{F}_{\Lambda}(\Pi q, -\mathcal{A}, \Pi \mathcal{U} \Pi; h, h_E) = \mathcal{F}_{\Lambda}(q, \mathcal{A}, \mathcal{U}; h, h_E). \quad (53)$$

Curvature changes sign, $F \mapsto -F$, preserving the source-free Maxwell equations $dF = 0$ and $d \star_g F = 0$.

VI. RELATIVE PHASE, PROPER TIME, AND INDEPENDENT TUNING

For a piecewise-smooth path $\gamma : [0, 1] \rightarrow M$, parallel transport solves

$$\dot{v} + \mathcal{A}(\dot{\gamma})C_1 v = 0. \quad (54)$$

In a global Abelian trivialization,

$$P_{\mathcal{A}}(\gamma) = \exp\left(-C_1 \int_{\gamma} \mathcal{A}\right). \quad (55)$$

For paths $\gamma_1, \gamma_2 : p \rightarrow q$, let $\ell_{12} = \gamma_1 * \overline{\gamma_2}$. Then

$$\text{Hol}_{\mathcal{A}}(\ell_{12}) = P_{\mathcal{A}}(\gamma_2)^{-1} P_{\mathcal{A}}(\gamma_1), \quad \Delta_{\text{phase}}(\gamma_1, \gamma_2) = \text{Hol}_{\mathcal{A}}(\ell_{12}) - I. \quad (56)$$

If the paths are timelike, define

$$\tau[\gamma] = \frac{1}{c} \int \sqrt{-g(\dot{\gamma}, \dot{\gamma})} d\lambda, \quad \Delta_{\text{time}}(\gamma_1, \gamma_2) = \tau[\gamma_1] - \tau[\gamma_2]. \quad (57)$$

The two differences have different codomains: $\Delta_{\text{phase}} \in \text{End}(E_p)$ and $\Delta_{\text{time}} \in \mathbb{R}$.

Lemma 7 (Fixed curvature, variable holonomy). *Let M carry a closed one-form η and a loop ℓ satisfying*

$$d\eta = 0, \quad \int_{\ell} \eta = 2\pi. \quad (58)$$

For a global $SO(2)$ connection form $\mathcal{A}_0$ on the trivial bundle, define

$$\mathcal{A}_a = \mathcal{A}_0 + a\eta, \quad a \in \mathbb{R}. \quad (59)$$

Then

$$d\mathcal{A}_a = d\mathcal{A}_0, \quad \text{Hol}_{\mathcal{A}_a}(\ell) = \text{Hol}_{\mathcal{A}_0}(\ell)e^{-2\pi a C_1}. \quad (60)$$

Proof. The curvature statement follows from $d\eta = 0$. The holonomy statement follows from Eq. (55), commutativity of $SO(2)$, and the normalization in Eq. (58). $\square$

When $\int_{\ell} \mathcal{A}_0 = 0$, varying a reaches every target element of $SO(2)$ without changing curvature.

VII. AN EXPLICIT LORENTZIAN WITNESS

Let

$$M = \mathbb{R}_t \times S^1_{\theta} \times \mathbb{R}_x \times \mathbb{R}_y, \quad (61)$$

and let η be the standard closed one-form on S^1 , normalized by $\int_{S^1} \eta = 2\pi$. Fix $c, \rho > 0$ and define

$$g = -c^2 dt^2 + \rho^2 \eta^2 + dx^2 + dy^2, \quad h = c^2 dt^2 + \rho^2 \eta^2 + dx^2 + dy^2. \quad (62)$$

Take the trivial bundle $E = M \times V$. Let $f \in C^2(\mathbb{R})$ satisfy $f' \neq 0$, put

$$u = t - \frac{x}{c}, \quad (63)$$

and define

$$\mathcal{A}_a = a\eta + f(u) dy. \quad (64)$$

Its curvature is independent of a :

$$F = d\mathcal{A}_a = f'(u) du \wedge dy. \quad (65)$$

Clearly $dF = 0$. The nonzero coordinate components are

$$F_{ty} = f'(u), \quad F_{xy} = -\frac{1}{c} f'(u). \quad (66)$$

Since the metric coefficients are constant,

$$\nabla_{\mu} F^{\mu y} = \partial_t F^{ty} + \partial_x F^{xy} = -\frac{1}{c^2} f''(u) + \frac{1}{c^2} f''(u) = 0, \quad (67)$$

and the other divergences vanish. Hence, in the standard differential-form description of source-free electromagnetism [9],

$$dF = 0, \quad d \star_g F = 0, \quad F \neq 0. \quad (68)$$

The dependence on $u = t - x/c$ is a propagating plane-wave profile.

Restrict the global bundle, connection, and any global section q to a finite trivializing cover, using identity transition maps. Every defect in Eqs. (43)–(45) vanishes, so

$$\mathcal{F}_{\Lambda} = 0. \quad (69)$$

Fix

$$p = (0, [0], 0, 0), \quad q = (T, [0], 0, 0), \quad T > \frac{2\pi\rho}{c}, \quad (70)$$

and define, for $0 \leq t \leq T$,

$$\gamma_0(t) = (t, [0], 0, 0), \quad \gamma_1(t) = \left(t, \left[\frac{2\pi t}{T}\right], 0, 0\right). \quad (71)$$

Both paths are timelike and join p to q . Their proper times are

$$\tau[\gamma_0] = T, \quad \tau[\gamma_1] = T\sqrt{1 - \left(\frac{2\pi\rho}{cT}\right)^2}, \quad (72)$$

hence

$$\Delta_{\text{time}}(\gamma_0, \gamma_1) \neq 0. \quad (73)$$

Let $\ell = \gamma_1 * \overline{\gamma_0}$. Along both paths $dy = 0$, while ℓ winds once around S^1 . Therefore

$$\int_{\ell} f(u) dy = 0, \quad \int_{\ell} \eta = 2\pi, \quad \text{Hol}_{\mathcal{A}_a}(\ell) = e^{-2\pi a C_1}. \quad (74)$$

Choose $a \notin \mathbb{Z}$. Then

$$\Delta_{\text{phase}}(\gamma_1, \gamma_0) \neq 0. \quad (75)$$

Varying a prescribes the holonomy arbitrarily in $SO(2)$ while leaving the nonzero Maxwell curvature in Eq. (65) unchanged.

Proof of Theorem 1. Equations (1)–(16) provide the common carrier, square symmetry, exact projective factorization, maximal equalizer, and infinite-dihedral dynamics. Proposition 4, Eqs. (19)–(22), and Proposition 5 give Fibonacci iteration, the sharp approximation scale, and the hyperbolic realization. Equations (32) and (33) place the completed-zeta reflection without extending its scope. Proposition 6 supplies exact local compatibility. The explicit data in Eqs. (61)–(75) satisfy the source-free Maxwell equations, exact gluing, nontrivial relative holonomy, and unequal proper time simultaneously. Lemma 7 proves independent curvature and holonomy tuning. $\square$

VIII. THE STRICT BOUNDARY

The construction contains several nonzero or exactly vanishing quantities, but they are not of one type:

$$\begin{aligned} \epsilon_{\Phi} &\in \mathbb{R}, \\ F &\in \Omega^2(M), \\ \Delta_{\text{phase}} &\in \text{End}(E_p), \\ \Delta_{\text{time}} &\in \mathbb{R}, \\ \mathcal{F}_{\Lambda} &\in \mathbb{R}_{\geq 0}. \end{aligned} \quad (76)$$

No equation in this paper identifies these objects or makes one a function of another. In particular:

- (a) the completed-zeta reflection supplies no statement about the location of zeta zeros;
- (b) $M_I \notin O(2)$, so the arithmetic reflection does not act as a gauge transformation or fiber isometry;
- (c) the common generator C_1 does not constrain the scalar Maxwell profile $f'(u)$;
- (d) exact gluing $\mathcal{F}_{\Lambda} = 0$ does not force zero curvature, zero holonomy, or equal proper time;
- (e) nonzero holonomy and unequal proper time are jointly readable but are not equated.

This boundary is part of the theorem. The result is coexistence under exact typing, not derivation of one canonical theory from another.

IX. CONCLUSION

The factorization

$$M_I C_1 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \quad (77)$$

is the smallest algebraic hinge in the construction. Its alternating projective action produces the Fibonacci matrices exactly, locates a maximal two-point meeting of distinct involutions, and fixes a hyperbolic axis whose length scale is governed by Φ . The same two-dimensional carrier admits a separate $SO(2)$ gauge office on a Lorentzian manifold, where exact gluing, nonzero Maxwell curvature, variable holonomy, and unequal proper time coexist in one explicit witness.

The point is exact and limited: these structures can inhabit one realization without being identified. Compatibility is proved by construction. Every stronger bridge remains open.

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